EX: 14 TO CAPTURE SAVE AND ANALYZE NETWORK TRAFFIC ON TCP/UDP/IP/HTTP/ ARP/DHCP/ICMP/DNS USING WIRESHARK TOOL

Aim: The aim of this experiment is to capture, save & analyze network traffic on TCP/UDP/IP/HTTP/ARP/DHCP/ICMP/ DNS Using wireshark tool.

Procedure:

Step 1: download & install wireshark from its official website.

Step 2: Launch wireshark & identify the network interface

Step 3: Before starting the capture, apply capture filters to focus on specific protocols

Capture only TCP: tcp, capture only UDP: udp
Capture only HTTP: tcp port 80, Capture only DNS: udp Port 53
capture only ICMP: icmp, capture only ARP: arp
capture DHCP: port 67 or 68

Step 4: click start to begin capturing live network traffic.

Step 5: capturing, use display filters to isolate & analyze specific protocols

Step 6: let capture run until you have collected enough packets for Analysis

Step 7: save captured traffic by navigating

Step 8: Review & analyze captured packets in detail, Look for Source & destination IP addresses, protocol types, packet lengths & Flags

Output:
No: packet number in capture sequence
Time: packet was captured
Source, destination
Protocol: TCP, UDP, HTTP, ARP, DNS
Length: size of packets in bytes

sample-output:

1. 0.00000 192.168.1.1 8.8.8.8 ICMP 56
Echo request

2. 0.002345 8.8.8.8 192.168.1.1 ICMP
64 Echo Reply

3. 0.003234 192.168.1.2 192.168.1.1
ARP 42 who has 192.168.1.1?

Result: Thus this experiment capture, sa
& analyze network traffic on TCP/UDP/(A
HTTP/ARP/DHCP/ICMP/DNS using
wireshark tool